GLUEING FINITE PRESENTATION SPACES NEAR A POINT OF A RETROCOMPACT OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4254-4284, following the statement at 4239-4252.

Editorial note. No omissions; statement differs from the closed-point variant in the left-hand colimit over U' and the retrocompact hypothesis.

Allowed standard prerequisites (editorial).

- Zariski glueing of finite presentation spaces. lemma-relative-glueing (same chapter, card L26319, cited at ~4255-4261). FP_{U cup W} = FP_U x_{FP_{U cap W}} FP_W.
- Local ring as colimit over affine neighbourhoods. lemma-descend-finite-presentation (same chapter, card L26286, cited at ~4271-4272). FP over Spec(O_{S,s}) = lim of FP over the affine neighbourhoods W.
- Limits of schemes with affine transitions. Limits, Lemma limits-lemma-directed-inverse-system-has-limit (cross-chapter, cited at ~4275-4277). V = lim (W cap U) is a limit of qcqs schemes.

Source reference. Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E8Z: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4154–4154. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SF: focused-seeds/geometry/S9-spaces-limits-glueing-near-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $U \subset S$ be a retrocompact open. Let $s \in S$ be a point in the complement of U. With $V = \mathrm{Spec}(\mathcal{O}_{S,s}) \cap U$ there is an equivalence of categories*

$$\mathrm{colim}_{s \in U' \supset U \text{ open}} FP_{U'} \longrightarrow FP_U \times_{FP_V} FP_{\mathrm{Spec}(\mathcal{O}_{S,s})}$$

where FP_T is the category of algebraic spaces of finite presentation over T.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $W \subset S$ be an open neighbourhood of s. By Lemma 0E8Z the functor

$$FP_{U \cup W} \longrightarrow FP_U \times_{FP_{U \cap W}} FP_W$$

is an equivalence of categories. We have $\mathcal{O}_{S,s} = \operatorname{colim} \mathcal{O}_W(W)$ where W runs over the affine open neighbourhoods of s. Hence $\operatorname{Spec}(\mathcal{O}_{S,s}) = \lim W$ where W runs over the affine open neighbourhoods of s. Thus the category of algebraic spaces of finite presentation over $\operatorname{Spec}(\mathcal{O}_{S,s})$ is the limit of the category of algebraic spaces of finite presentation over W where W runs over the affine open neighbourhoods of s, see Lemma 07SK. For every affine open $s \in W$ we see that $U \cap W$ is quasi-compact as $U \to S$ is quasi-compact. Hence $V = \lim W \cap U$ is a limit of quasi-compact and quasi-separated schemes (see Limits, Lemma 07SF). Thus also the category of algebraic spaces of finite presentation over V is the limit of the categories of algebraic spaces of finite presentation over $W \cap U$ where W runs over the affine open neighbourhoods of s. The lemma follows formally from a combination of these results. $\square$